

Introduction to Atmospheric and Climate Sciences EES 202 2024 monsoon semester

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Outline of Course Content

• **EES202: Introduction to atmospheric and climate sciences [Cr: 3, Lc:2, Tt:1]**

• **Course Outline**

- Composition of Earth's Atmosphere, Mass of the Atmosphere, Temperature
- Structure and Atmospheric Regions, Moisture in the atmosphere
- Scale height, potential temperature, Lapse rates, geopotential, hydrostatic balance, buoyancy frequency, geostrophic balance, Atmospheric Dynamics, Fundamental Forces of the atmosphere, Stability, Turbulence and Vertical Transport, General Circulation, Water (vapour, clouds and rain)
- Evolution of the Atmosphere and the climate system, Radiation Characteristics, Distribution of Insolation, Greenhouse Effect, Scattering by Aerosols and Clouds
- Introduction to climate and climate change, Natural and anthropogenic climate
- Change, Climate Change in Deep Time: Snowball Earth, Icehouses and Greenhouses
- Stratospheric Chemistry, Ozone Observations, Chapman Model, Catalytic Loss
- Cycles, Antarctic Ozone Hole, Montreal Protocol
- Tropospheric Chemistry, Sources and Sinks of Major Atmospheric Constituents
- and Oxidants, Air Quality, Smog, Acid rain and Air Pollution to Climate Change

Main text book • **D. J. Jacob, Introduction to Atmospheric Chemistry, Princeton University Press; For Climate Science topics: IPCC report: Summary for Policy Makers**

Grading Criteria: EES 202; 2024 Monsoon semester

Total Marks: **100**

Mid Sem Exam I : **20**

Mid Sem exam II : **20**

Numerical Assignment/Quiz: **7**

End Sem Exam: **50**

Attendance: 3 Marks ($\geq 90\%$:3/3; 80-89% 2/3; 60-79% 1/3; less than 60%= 0/3); attendance includes tutorial sessions

Tutorial and mentoring sessions for interactive discussions and problem solving on Fridays

Further suggested reading:

Additional reading • J. H. Seinfeld and S.N. Pandis, Atmospheric Chemistry and Physics: From Air Pollution to Climate Change, Wiley-Interscience (1998).

• B.J. Finlayson-Pitts and J.N. Pitts Jr., Chemistry of the Upper and Lower

• Atmosphere, Academic Press (2000); J. R. Holton, An introduction to Dynamic Meteorology, Elsevier Academic Press (2004), R. Ruddiman, Earth's climate, Past Present Future, W. H. Freeman (2000), Paperback, 465 pages, ISBN 0716737418 / 9780716737414

WHAT IS THE ATMOSPHERE?

- Gaseous envelope surrounding the Earth
- Mixture of gases, also contains suspended solid and liquid particles (*aerosols*)

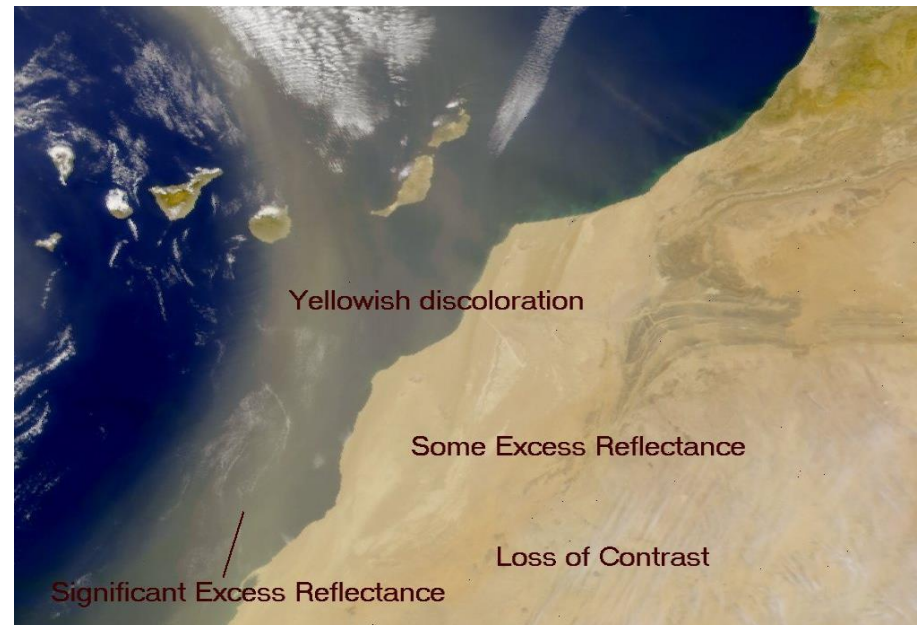
Aerosol = dispersed condensed phase suspended in a gas

Aerosols are the “visible” components of the atmosphere

Pollution haze over East Coast

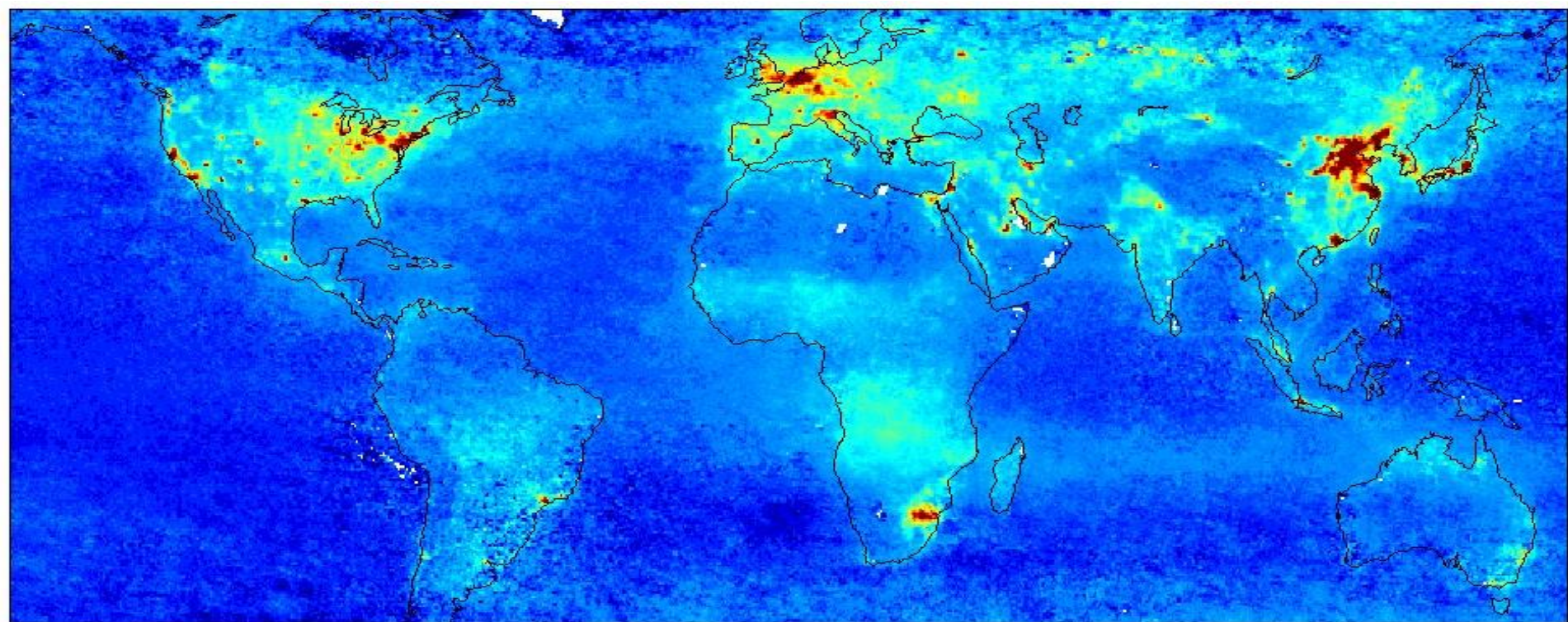


Dust off West Africa



ATMOSPHERIC GASES ARE “VISIBLE” TOO... IF YOU LOOK IN THE UV OR IR

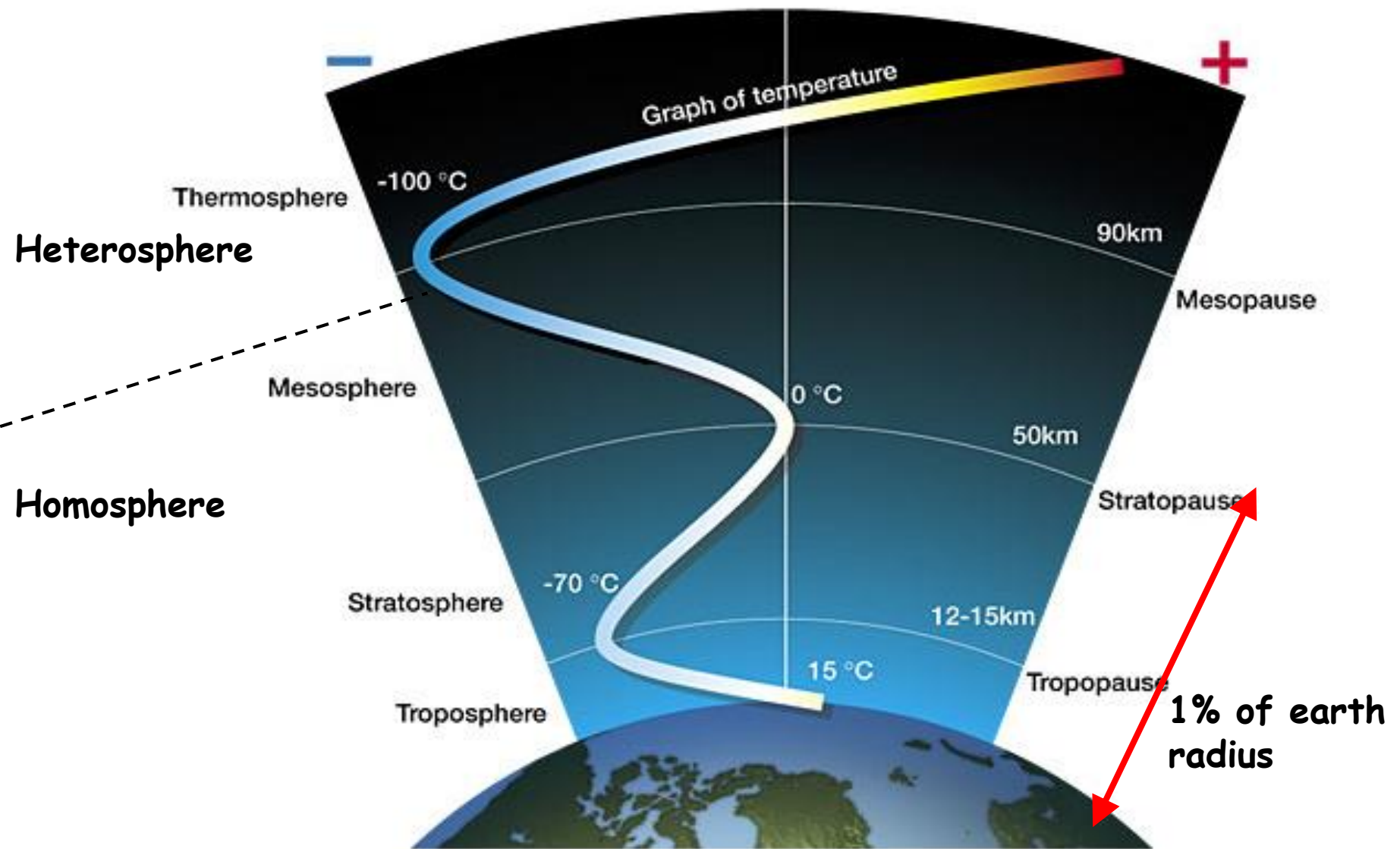
NO₂ Columns Observed from the SCIAMACHY Satellite Instrument



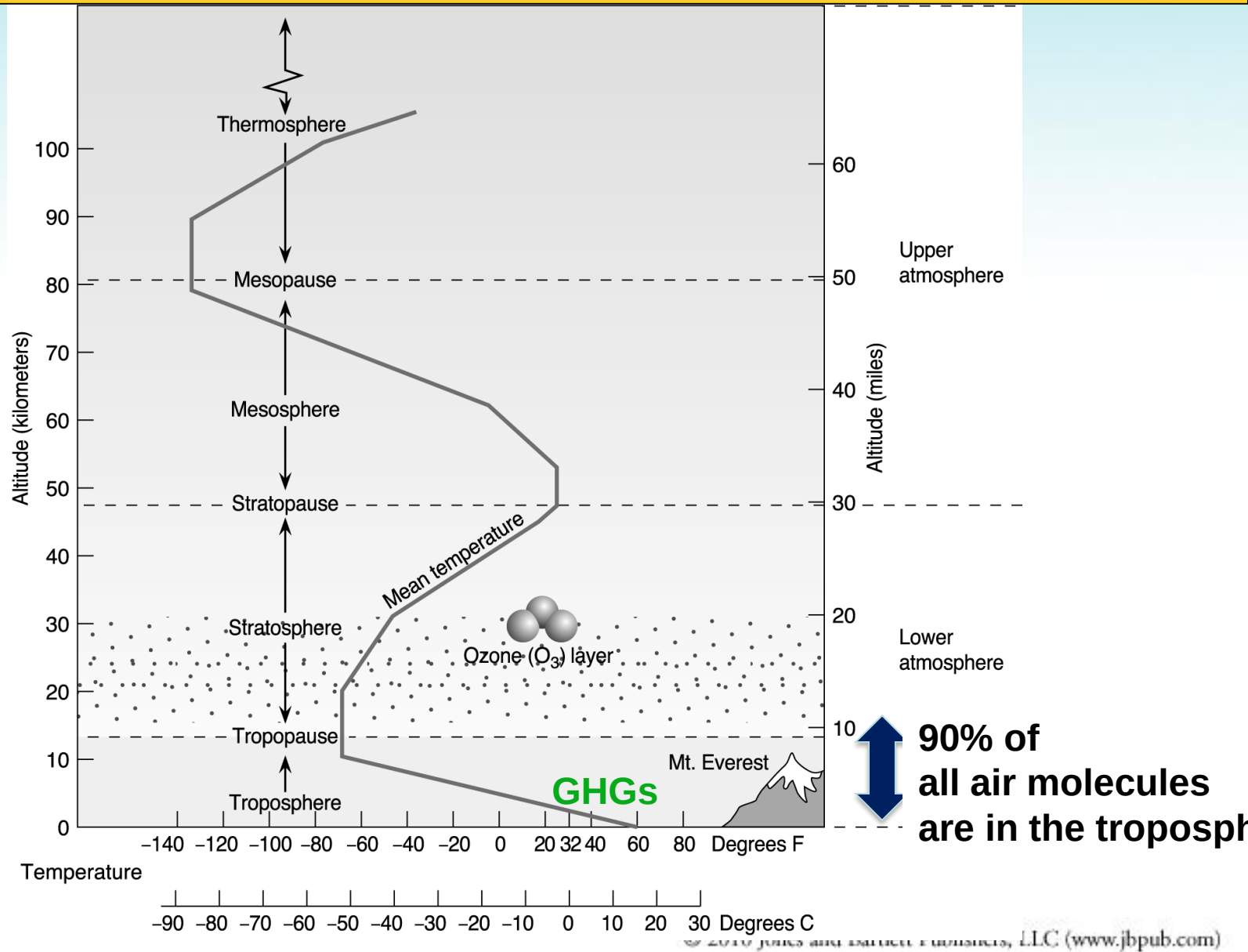
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SCIAMACHY Tropospheric NO₂ (10^{15} molec cm⁻²)

The layers of the atmosphere

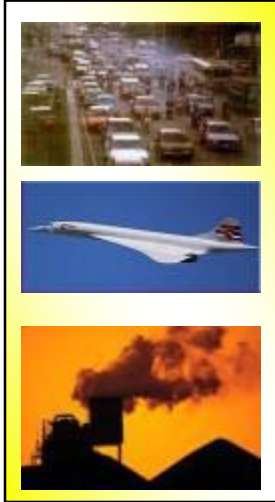


The atmosphere as a chemical reactor (T, P and radiation changes)

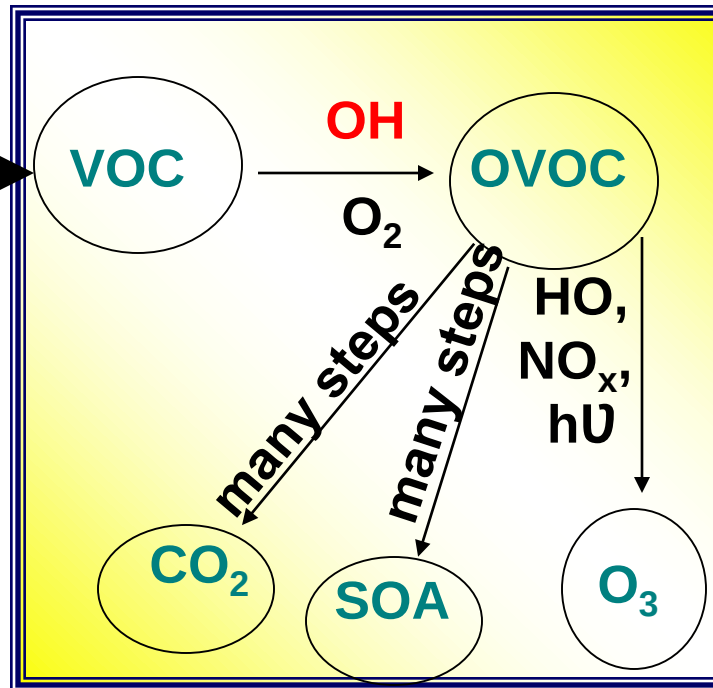


Atmosphere is Modulator of Climate and AQ: Emissions-Oxidation/Atmospheric Chemistry -Air Quality-Climate

Emissions



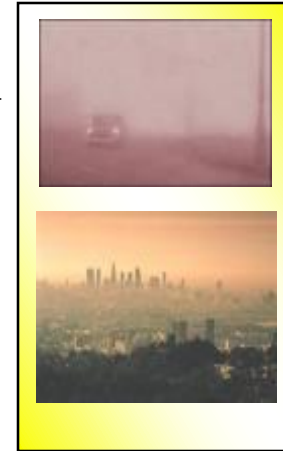
Role of VOC & OH oxidation



Primary Emissions (Gases and Aerosol)

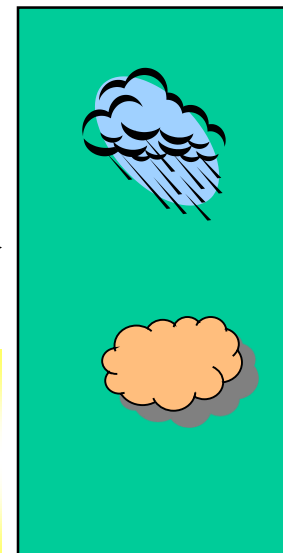
Approx. 1300 TgC / yr from NMHCs alone
(Goldstein and Galbally, Env. Sci. & Tech, 2007)

Impacts



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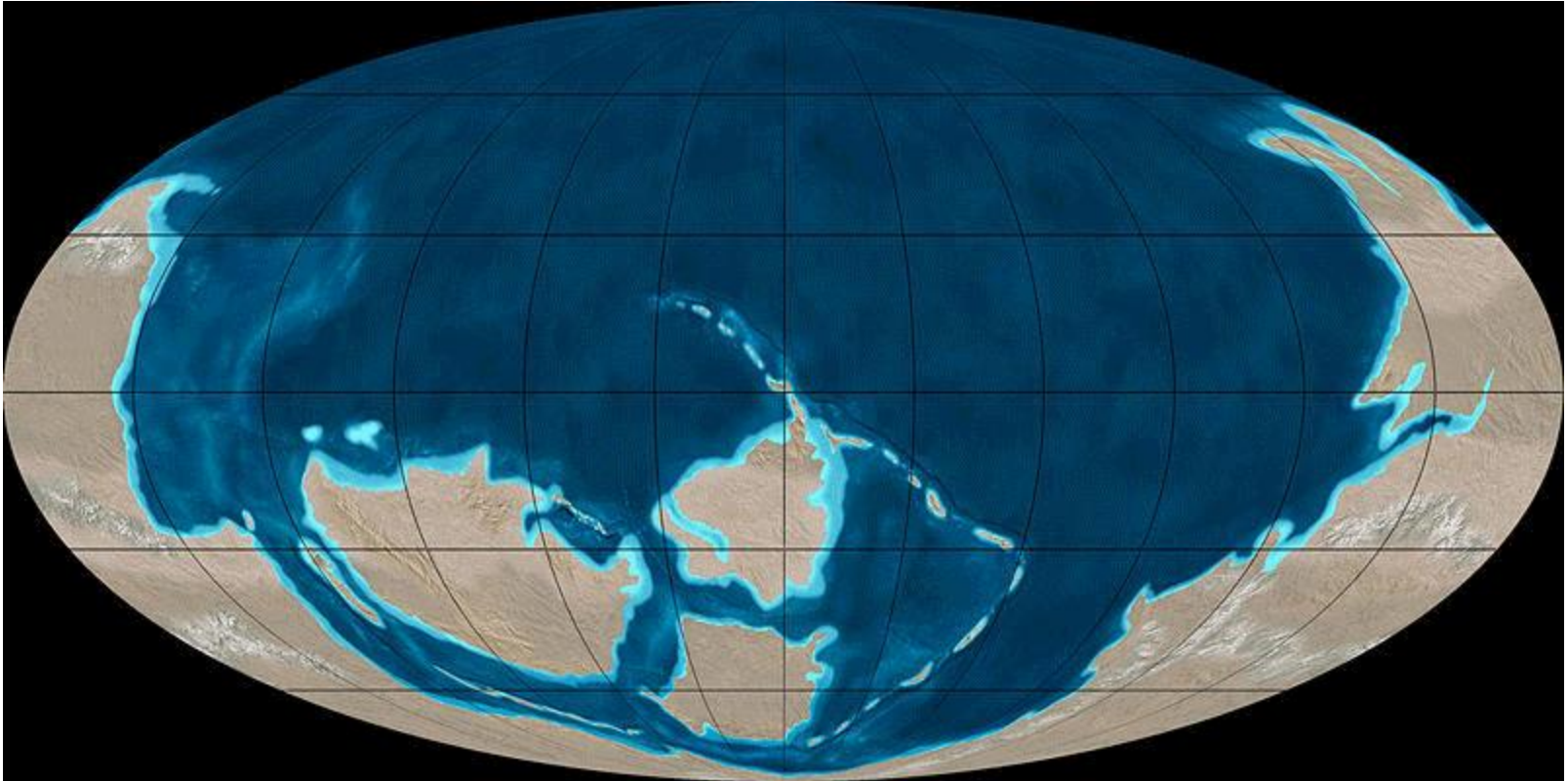


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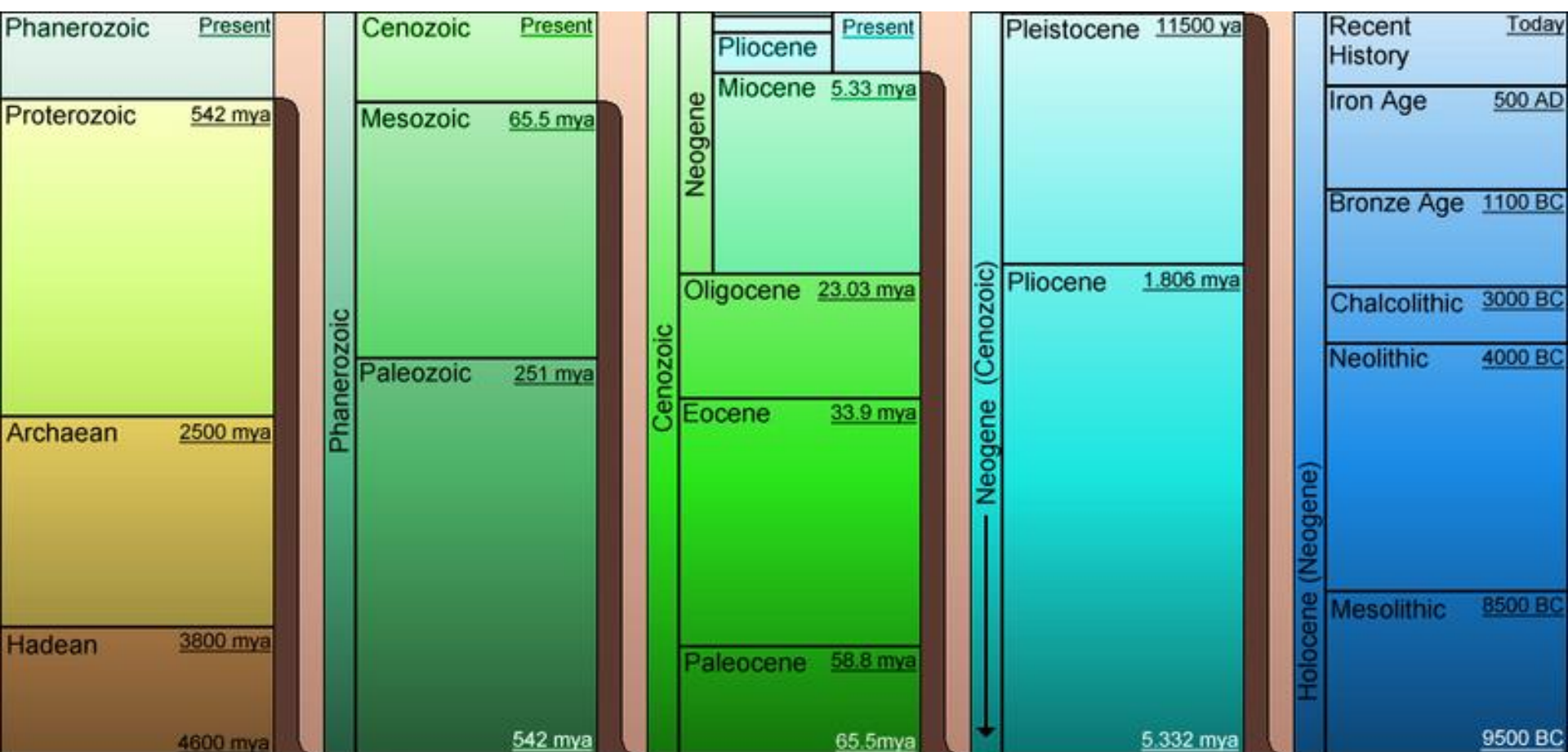
**HOW DID THE EARTH'S ATMOSPHERE COME
ABOUT, LETS SEE WHAT CHANGES THE EARTH
HAS UNDERGONE IN THE PAST**

Movement of Tectonic Plates In the Earth's History



Source: Wikipedia

Chronology of Different Geological Time periods In the Earth's History



If you made a Geological Timescale with one year for every millimeter, the entire list would be about 4,600 kilometers (about 2,858 miles) in length.

Human-like primates showed up in the last 2 or 3 kilometers (the last mile or two) of the list.

By comparison, dinosaurs lived for about 160 kilometers of that list (about 100 miles).

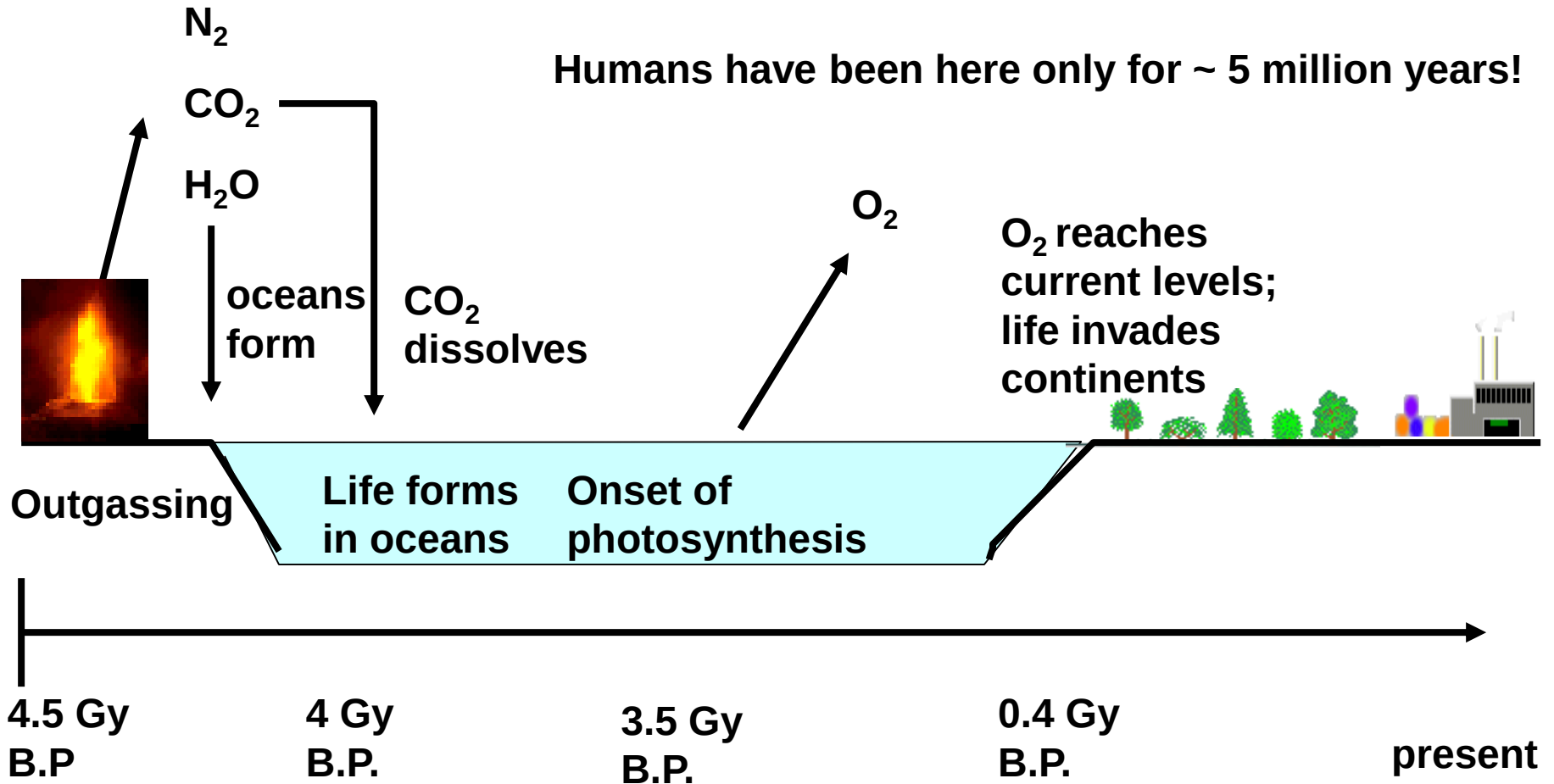
Modern humans would begin to appear about 10 to 15 meters (about 32 to 49 feet) from the end of that list (depending on what you consider "modern").

A 30 year old person would have only been on that list for the last 3 centimeters (about 1.2 inches).

Evolution of Earth's Atmosphere: Salient Features

- **The Earth was formed ~ 4.5 billion yrs ago by gravitational accretion of a cloud of gases and dust**
- **The Earth system (including solid Earth and atmosphere) is an assemblage of mainly 92 natural elements (O, Fe, Si, H, N, C, S)**
- **Volcanic outgassing transferred material from the deep Earth to the atmosphere**
- **Outgassed material was mainly H₂O (~95%), CO₂, N₂ and sulphur gases; No O₂ !**
- **Gravity prevents escape of atoms (except for H and He for which loss is negligible)**
- **Outgassed water precipitated to form oceans into which CO₂ and sulphur gases dissolved over time; N₂ remained in the atmosphere**

Evolution & History of Earth's Atmosphere

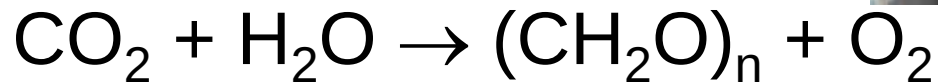


Source: Daniel Jacob, Harvard Univ

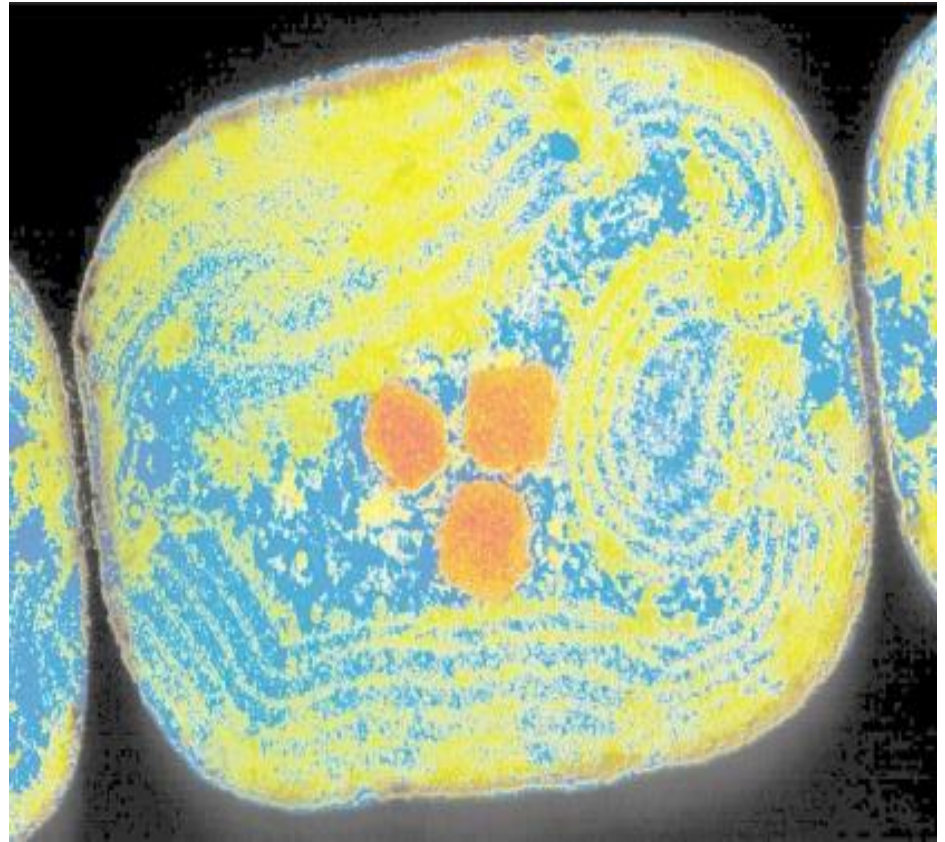
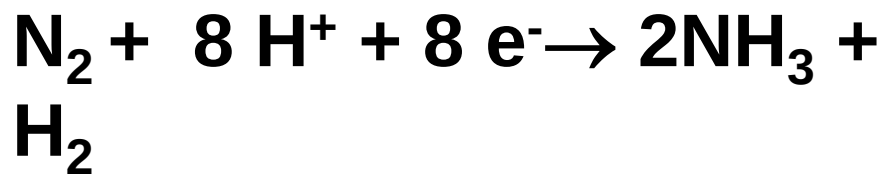
Cyanobacteria:

The first organisms making our planet habitable

Photosynthesis:



Nitrogen fixation:



3.5 Billion years ago some organisms developed the ability to convert CO_2 to organic carbon by photosynthesis, which also released O_2

Size/Mass of the atmosphere

Radius of Earth = 6400 km

Area: $A = 5.1 \times 10^{14} \text{ m}^2$

Global mean pressure at surface $P_s = 984 \text{ hPa}$; $1 \text{ hPa} = 100 \text{ Pa}$

$$\rightarrow M_a \times g = 4 \times 3.142 \times R^2 \times P_s$$

$$\rightarrow \text{Mass of the atmosphere: } 5.2 \times 10^{18} \text{ kg}$$

$> 10^{44}$ molecules

1. MEASURES OF ATMOSPHERIC COMPOSITION

1.1 Mixing ratio or mole fraction C_x [mol mol⁻¹]

$$C_x = \frac{\# \text{ moles of X}}{\text{mole of air}}$$

remains constant when air density changes

⇒ robust measure of atmospheric composition

e.g. of room air filled balloon

- Air also contains variable H₂O vapor (10⁻⁶-10⁻² mol mol⁻¹) and aerosol particles (diff. sweating)

- H₂ = 500 ppb

- N₂O = 320 ppb

- Trace gas abundance levels :

1 ppmv = 1x10⁻⁶ mol mol⁻¹

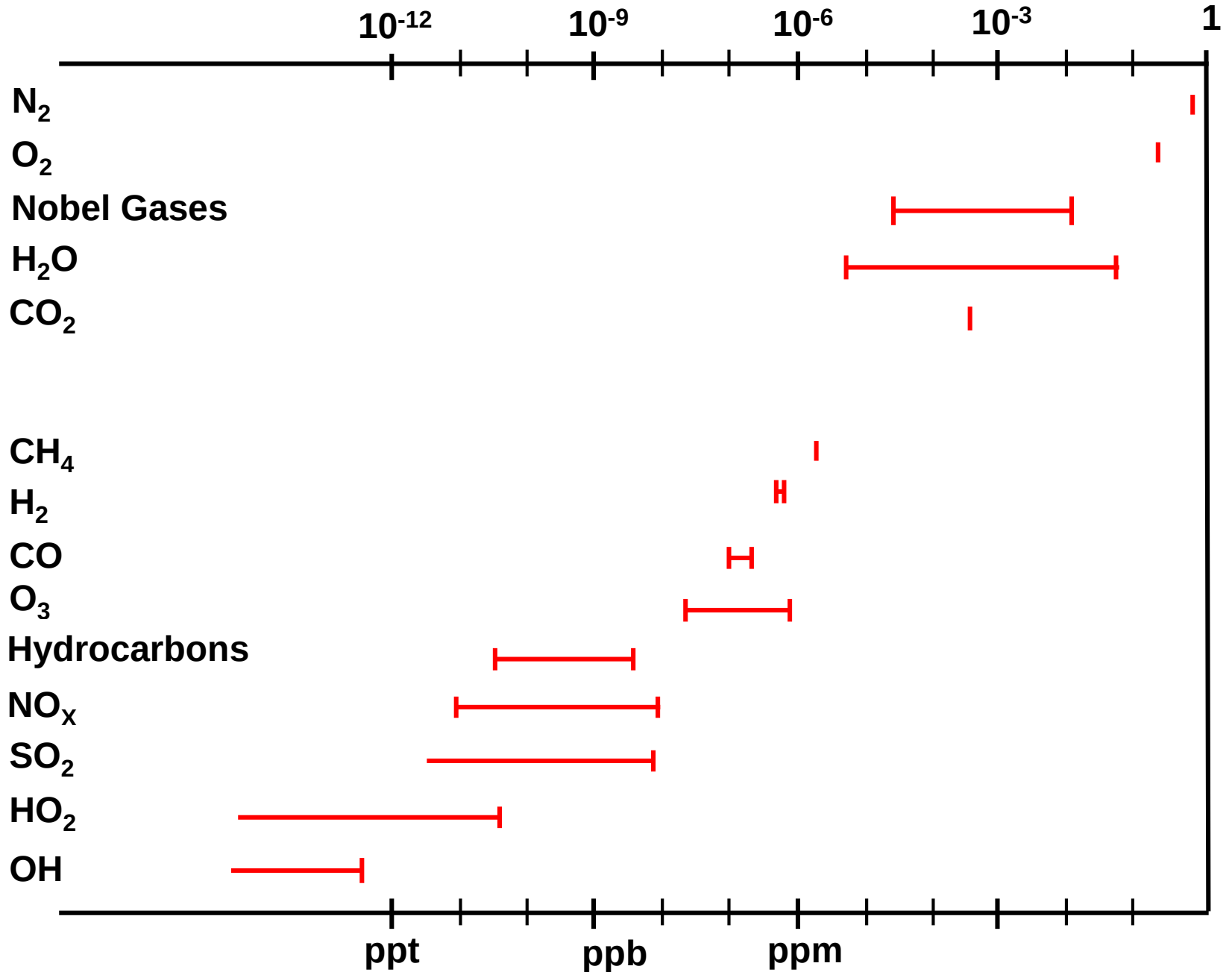
1 ppbv = 1x10⁻⁹ mol mol⁻¹

1 pptv = 1x10⁻¹² mol mol⁻¹

GAS	MIXING RATIO (dry air) [mol mol ⁻¹]
Nitrogen (N ₂)	0.78
Oxygen (O ₂)	0.21
Argon (Ar)	0.0093
Carbon dioxide (CO ₂)	400 x10 ⁻⁶
Neon (Ne)	18x10 ⁻⁶
Ozone (O ₃)	(0.01-10)x10 ⁻⁶
Helium (He)	5.2x10 ⁻⁶
Methane (CH ₄)	1.8x10 ⁻⁶
Krypton (Kr)	1.1x10 ⁻⁶

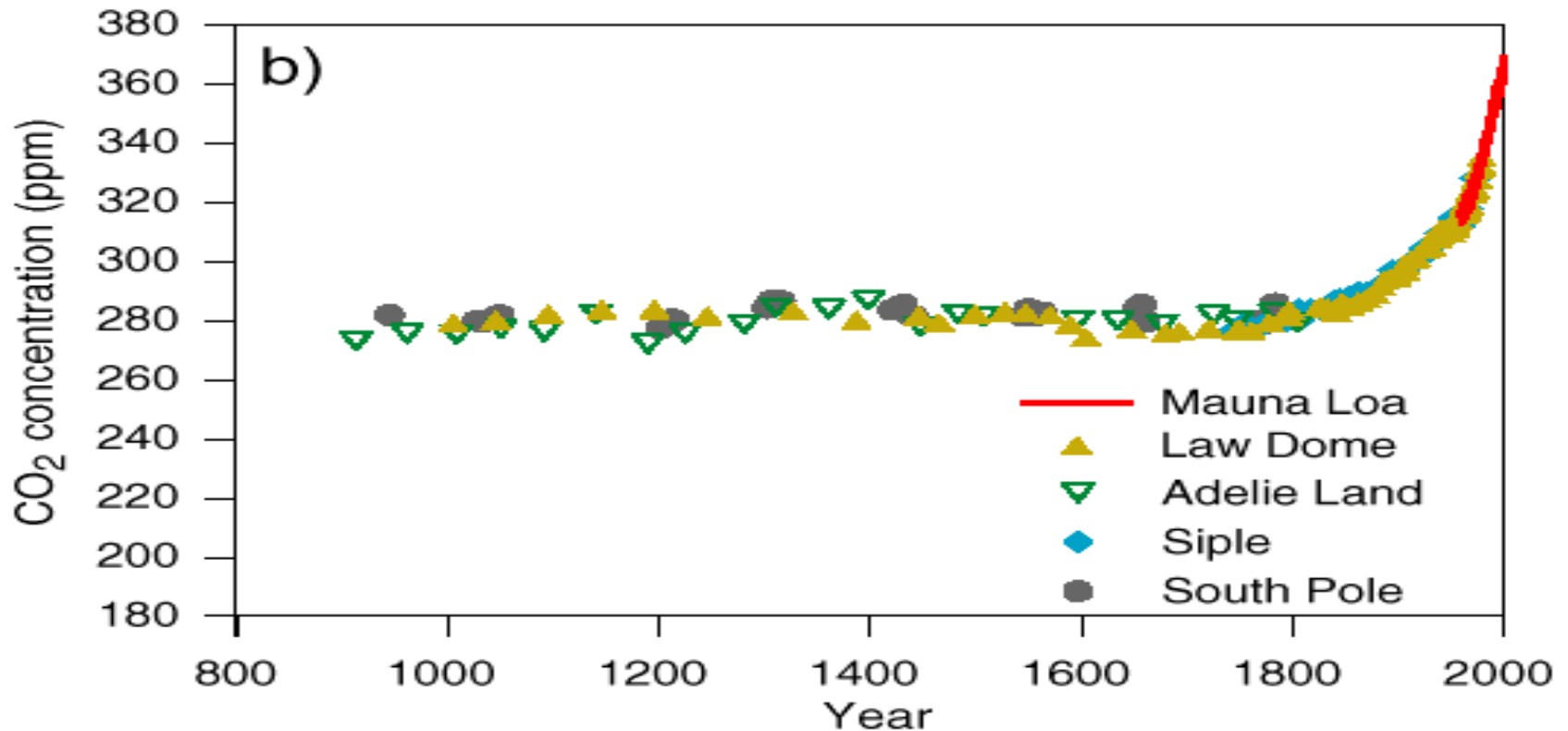
Trace
gases

Mixing Ratios of Atmospheric Trace Gases



ATMOSPHERIC CO₂ INCREASE OVER PAST 1000 YEARS

Intergovernmental Panel on Climate Change (IPCC), 2007

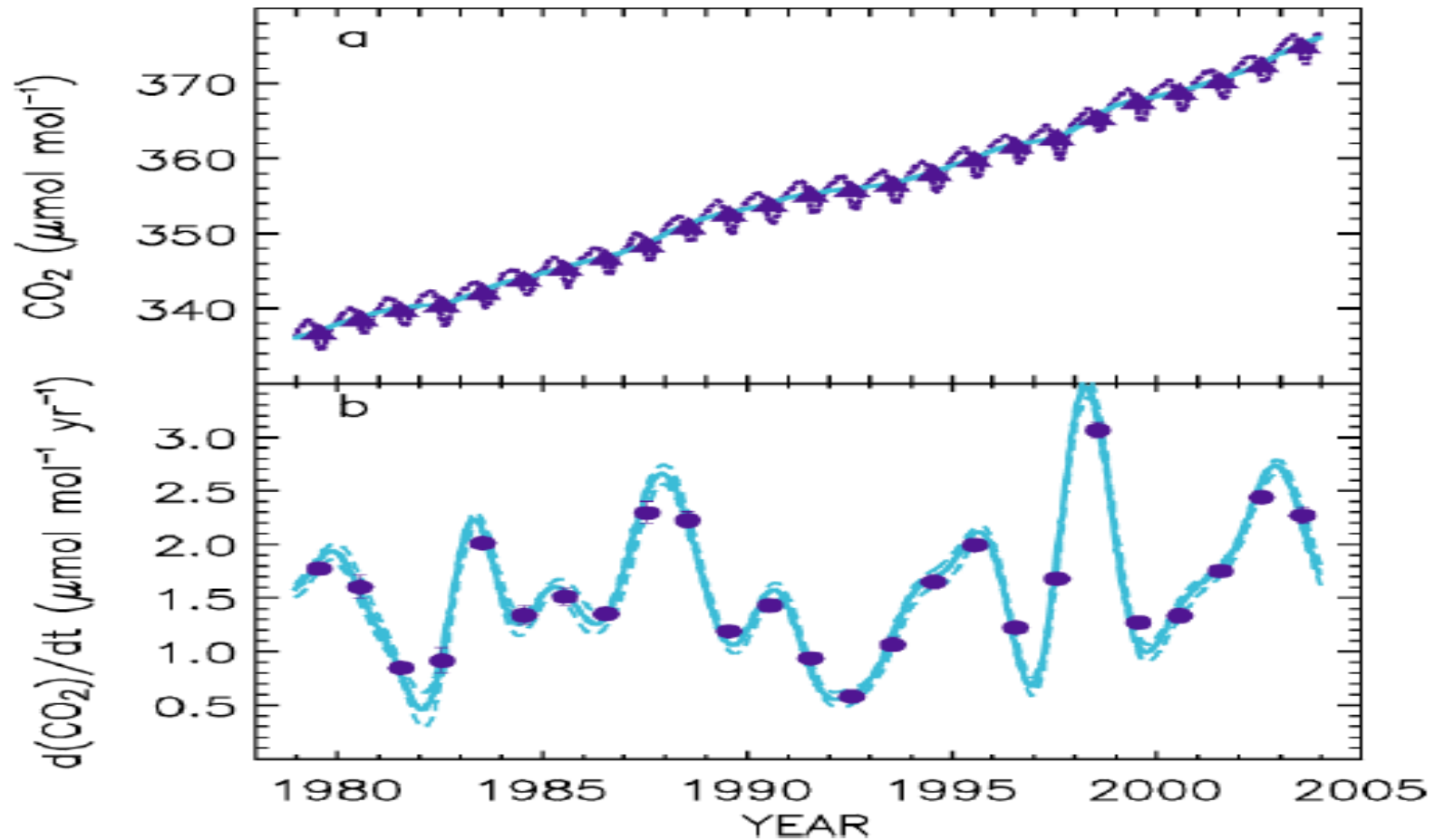


Abundance level units: parts per million (ppm)
number of CO₂ molecules per 10⁶ molecules of air

CO₂ CONCENTRATION IS MEASURED AS MIXING RATIO

ATMOSPHERIC CO₂ TREND OVER PAST 25 YEARS

IPCC [2007]



$\mu\text{mol mol}^{-1}$ is the proper SI unit; ppm, ppmv are customary units

Problem 1.



What is the mean molecular weight of dry air ?

$$M_a = \sum_i C_i M_i$$

Answer: $28.96 \times 10^{-3} \text{ kg/mol}$

1.2 Number density n_X [molecules cm⁻³]

$$n_X = \frac{\text{\# molecules of X}}{\text{unit volume of air}}$$

Proper measure for
• reaction rates ($X + Y \longrightarrow P + Q$)

Column concentration $\Omega_X = \int_0^{\infty} n_X(z) dz$ Proper measure for absorption or scattering of radiation by atmosphere

n_X and C_X are related by **the ideal gas law**:

$$n_X = n_a C_X = \frac{A_v P}{RT} C_X$$

Also define the mass concentration (g cm⁻³):

Atmos. Aerosol: 0.01 – 10 μm

$$\rho_X = \frac{\text{mass of X}}{\text{unit volume of air}} = \frac{M_X n_X}{A_v}$$

n_a = air density

A_v = Avogadro's number

P = pressure

R = Gas constant

$= A_v k$ k = Boltzmann const

T = temperature

M_X = molecular mass of X

n_X is not conserved when

P or T changes

Problem 2.



Calculate the number densities of air and CH_4 at sea level for $P= 1013 \text{ hPa}$, $T= 0^\circ \text{ C}$)

Be careful with units!

Answer: $n_{\text{air}} = 2.69 \times 10^{25} \text{ molecules/ m}^3$

$n_{\text{ch4}} = 4.8 \times 10^{19} \text{ molecules/ m}^3$

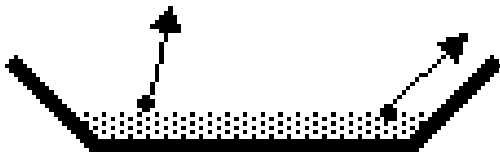
Units of Partial Pressure: Atmospheric context

- Partial pressure P_x of a gas X in a mixture of gases of total pressure P is the pressure that would be exerted by molecules of X if all other gases were removed from the mixture: Dalton's law states that $P_x = C_x \cdot P$
- From the ideal gas law $P_x = (n_x / A_v) \cdot R \cdot T$
- Partial pressure relates to the collision frequency of gas molecules with surfaces and therefore determines the exchange rate of molecules between a gas phase and a coexistent condensed phase
- This unit is of interest for gases that undergo phase changes in the atmosphere e.g. water vapour

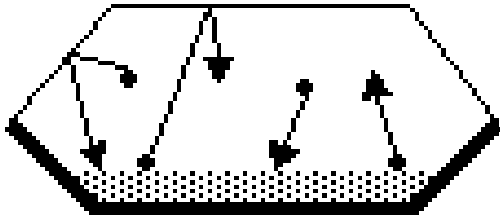
1.3 Partial pressure P_x [Pa]

Dalton's law: $P_X = C_X P$ Proper measure for phase change
(such as condensation of water vapor)

Evaporation of liquid water from a pan:



No lid: water molecules escape from pan to atmosphere (evaporation)



Add a lid:

- escaping water molecules collide on lid and return to surface; collision rate measures P_{H_2O}
- eventually, flux escaping = flux returning : saturation ($P_{H_2O, SAT}$)
- cloud formation in atmosphere requires $P_{H_2O} > P_{H_2O, SAT}$
- $T \nearrow \Leftrightarrow P_{H_2O, SAT} \nearrow$

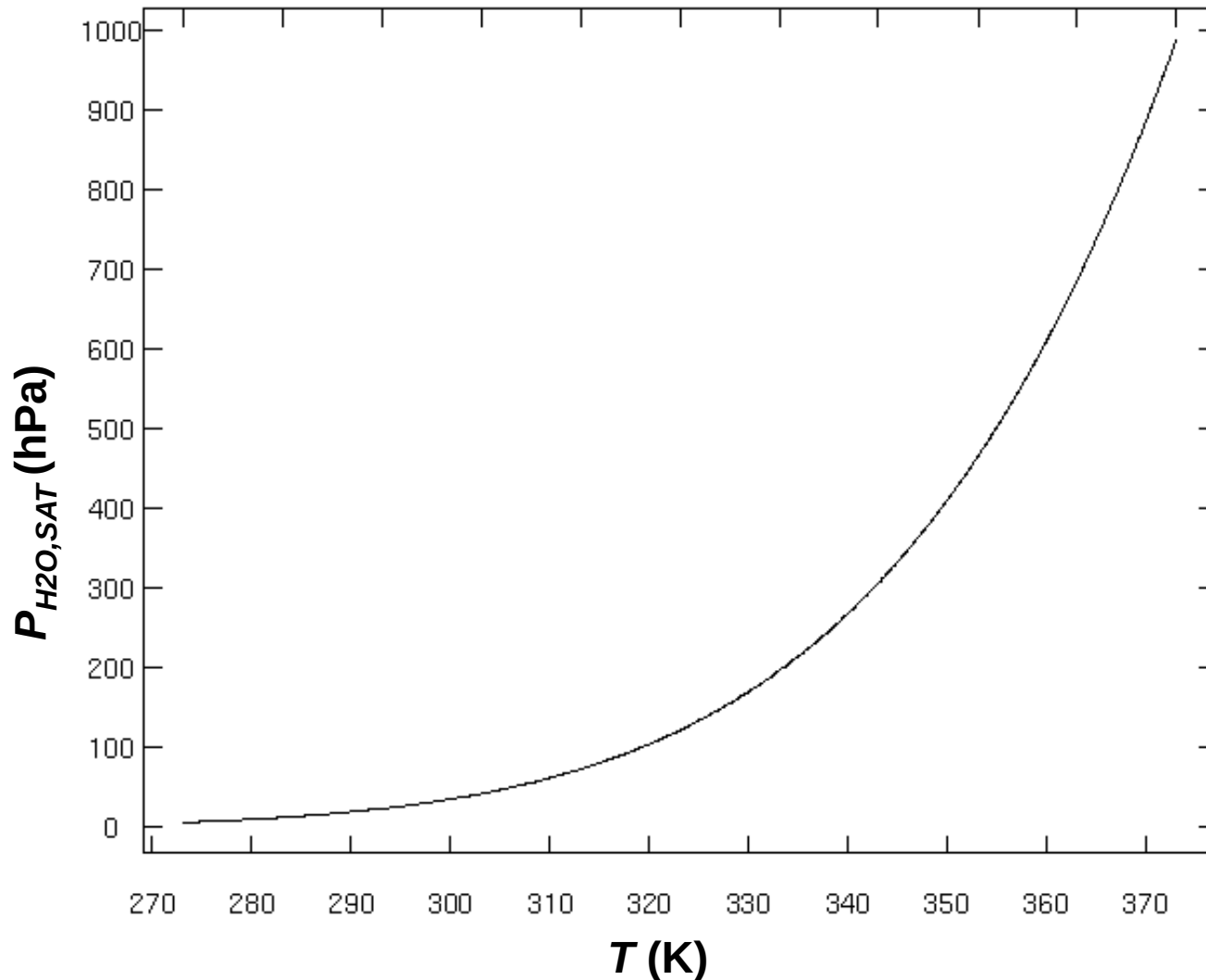
CLAUSIUS-CLAPEYRON EQUATION: $P_{H_2O, SAT} = f(T)$

$$P_{H_2O, SAT} = A \exp\left[B\left(\frac{1}{T} - \frac{1}{T_o}\right)\right]$$

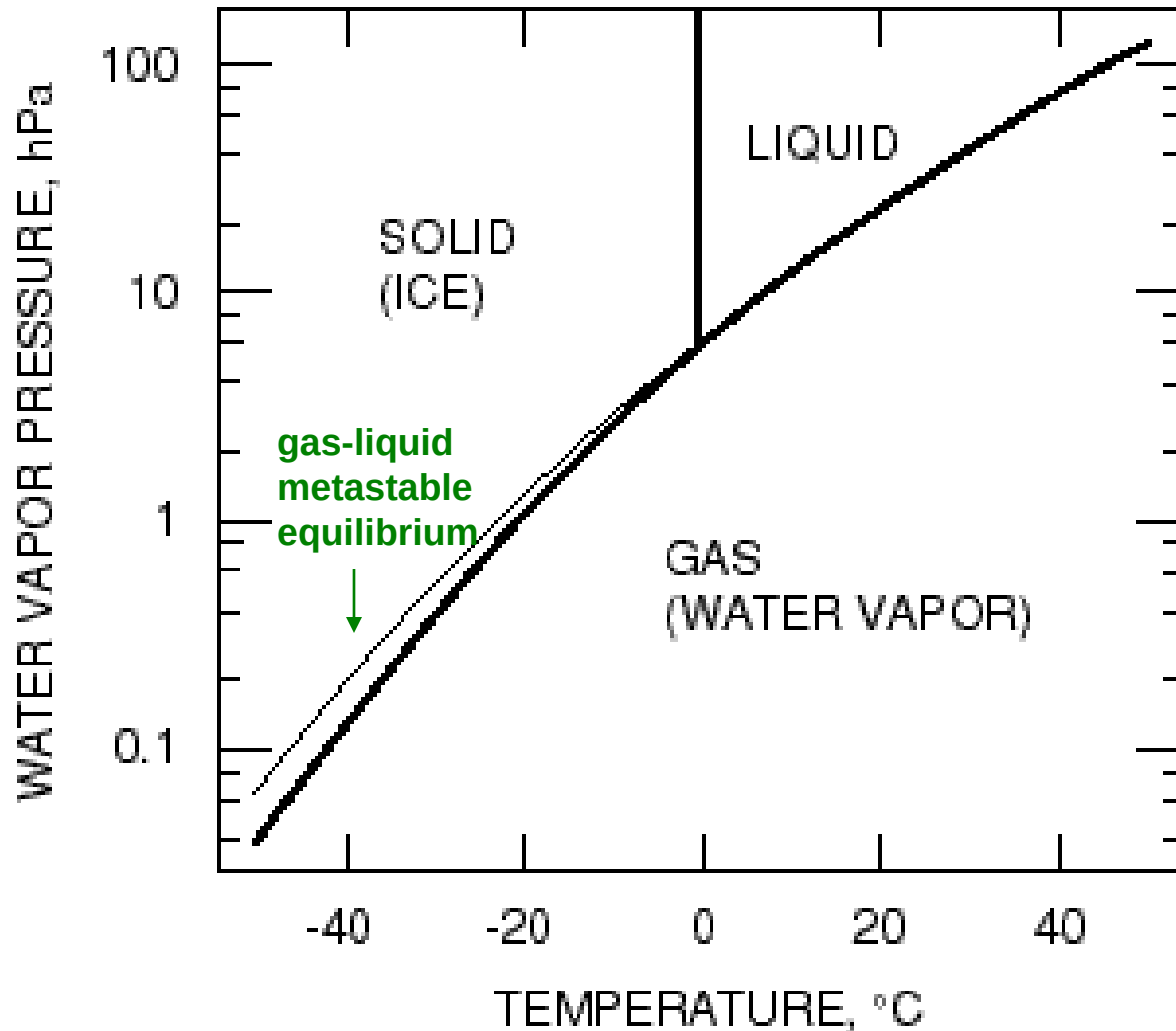
$$A = 6.11 \text{ hPa}$$

$$B = -5310 \text{ K}$$

$$T_o = 273 \text{ K}$$



PHASE DIAGRAM FOR WATER



$$\text{Relative humidity (\%)} = 100(P_{H_2O}/P_{H_2O,SAT})$$

Dew point: Temperature T_d such that $P_{H_2O} = P_{H_2O,SAT}(T_d)$

COMPARING THE ATMOSPHERES OF EARTH, VENUS, AND MARS

	Venus	Earth	Mars
Radius (km)	6100	6400	3400
Surface pressure (atm)	91	1	0.007
CO ₂ (mol/mol)	0.96	3x10 ⁻⁴	0.95
N ₂ (mol/mol)	3.4x10 ⁻²	0.78	2.7x10 ⁻²
O ₂ (mol/mol)	6.9x10 ⁻⁵	0.21	1.3x10 ⁻³
H ₂ O (mol/mol)	3x10 ⁻³	1x10 ⁻²	3x10 ⁻⁴

Source:
Finlayson and Pitts

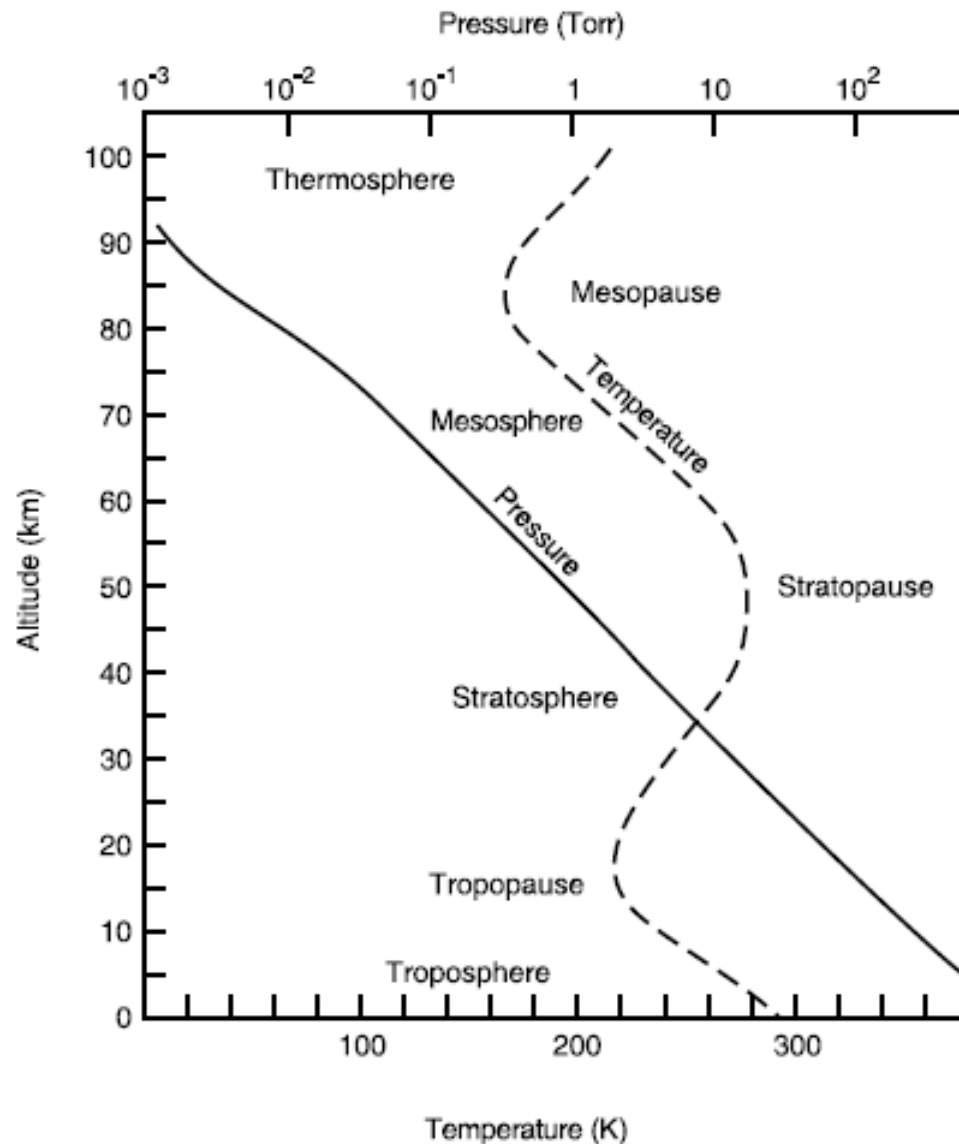


FIGURE 1.1 Typical variation of temperature with altitude at mid-latitudes as a basis for the divisions of the atmosphere into various regions. Also shown is the variation of total pressure (in Torr) with altitude (top scale, base 10 logarithms) where 1 standard atmosphere = 760 Torr.